

ELECTRONIC ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

Instruction: Using Microsoft Word/WordPad/Notepad, type Student's name, Student ID number, type Chapter title and applicable Question Number for Assignment, Mid-Term Exam or Final Exam (as listed under Brightspace, Quizzes) before typing Solution below. One separate file for each Assignment/Mid-Term/Final Exam Question.

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Assignment – Chapter title – applicable Question number:

Assignment – AM (3%) – Question 1

Solution:

(a)

$$\begin{aligned} X_1^T Y_1 &= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} [1 \quad 1 \quad -1 \quad -1] & X_2^T Y_2 &= \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} [1 \quad -1 \quad -1 \quad 1] \\ &= \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix} & &= \begin{bmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} W &= X_1^T Y_1 + X_2^T Y_2 \\ &= \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix} + \begin{bmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 & -2 & 0 \\ 0 & 2 & 0 & -2 \\ 2 & 0 & -2 & 0 \\ 0 & 2 & 0 & -2 \end{bmatrix} \end{aligned}$$

$$(b) X_t = [1 \quad -1 \quad 1 \quad 1]$$

$$G_h[X_t W] = G_h[[1 \quad -1 \quad 1 \quad 1] W]$$

$$= G_h[[4 \quad 0 \quad -4 \quad 0]]$$

$$= [1 \quad 1 \quad -1 \quad 1]$$

$$= Y_t$$

$$G_h[Y_t W^T] = G_h[[1 \quad 1 \quad -1 \quad 1] W^T]$$

$$= G_h[[4 \quad 0 \quad 4 \quad 0]]$$

$$= [1 \quad 1 \quad 1 \quad 1]$$

$$= X_1$$

$$G_h[X_1 W] = Y_1$$

$$G_h[Y_1 W^T] = X_1$$

The steady-state recall pattern pair is X_1 and Y_1 .